Long Division

Divide, multiply, take away, bring down. Repeat.

7,488 ÷ 24

74 ÷ 24 → 3	3 × 24 = 72	74 − 72 = 2, bring down 8: 28
28 ÷ 24 → 1	1 × 24 = 24	28 − 24 = 4, bring down 8: 48
48 ÷ 24 → 2	2 × 24 = 48	48 − 48 = 0

7,488 ÷ 24 = 312 Each step is one digit of the answer.

Use easy parts to guess each digit (packet 78)

How many 24s in 74? 24 × 3 = 72 fits, 24 × 4 = 96 is too much. So 3.
If what is left at the end is not 0, write it as R: 500 ÷ 24 = 20 R 20.

Your turn!

936 ÷ 12 = 78

For teachers

Goal: divide multi-digit numbers with the standard algorithm (6.NS.2). Say “Divide, multiply, take away, bring down.” Kids can still use packet 78's easy parts to find each digit.

Long Division: Practice

Divide, multiply, take away, bring down.

Part 1 One-digit divisors

Work under the number.

1) $4\overline{)852}$ = 213

2) $6\overline{)1{,}356}$ = 226

3) $5\overline{)2{,}745}$ = 549

4) $3\overline{)918}$ = 306

5) $8\overline{)4{,}096}$ = 512

6) $7\overline{)3{,}542}$ = 506

7) $9\overline{)1{,}872}$ = 208

8) $5\overline{)6{,}215}$ = 1,243

9) $2\overline{)2{,}334}$ = 1,167

Part 2 Two-digit divisors

Use easy parts to find each digit.

10) $24\overline{)7{,}488}$ = 312

11) $31\overline{)5{,}208}$ = 168

12) $23\overline{)9{,}177}$ = 399

Long Division: Practice (continued)

Part 2 Two-digit divisors (continued)

13) 384) 18 | 6,912

14) 198) 42 | 8,316

15) 187) 25 | 4,675

16) 243) 14 | 3,402

17) 210) 47 | 9,870

18) 95) 63 | 5,985

Part 3 Divide No help this time.

19) 3,920 ÷ 16 = 245

20) 3,744 ÷ 32 = 117

21) 8,316 ÷ 27 = 308

22) 7,290 ÷ 45 = 162

23) 6,838 ÷ 13 = 526

24) 8,178 ÷ 58 = 141

Long Division: Practice (continued)

Part 3 Divide (continued)

25) 9,093 ÷ 21 = 433

26) 7,452 ÷ 36 = 207

27) 7,296 ÷ 19 = 384

28) 8,000 ÷ 64 = 125

29) 8,848 ÷ 28 = 316

30) 8,823 ÷ 51 = 173

Part 4 What is left? No help this time.

31) 2,000 ÷ 37 = 54 R 2

32) 5,431 ÷ 16 = 339 R 7

33) 7,777 ÷ 25 = 311 R 2

34) 3,210 ÷ 43 = 74 R 28

35) 9,001 ÷ 12 = 750 R 1

36) 6,543 ÷ 58 = 112 R 47

Long Division: Practice (continued)

Part 4 What is left? (continued)

37) 4,444 ÷ 33 = 134 R 22

38) 8,123 ÷ 21 = 386 R 17

Part 5 More long division

39) 13) 1,326 = 102

40) 56) 12,320 = 220

41) 30) 5,490 = 183

42) 58) 8,874 = 153

43) 14) 2,044 = 146

44) 35) 10,360 = 296

45) 13) 598 = 46

46) 27) 6,183 = 229

47) 48) 12,288 = 256

Long Division: Practice (continued)

Part 6 More dividing

48) 6,260 ÷ 20 = 313

49) 15,120 ÷ 45 = 336

50) 8,004 ÷ 69 = 116

51) 17,520 ÷ 73 = 240

52) 1,386 ÷ 33 = 42

53) 3,904 ÷ 16 = 244

54) 8,225 ÷ 25 = 329

55) 1,620 ÷ 18 = 90

56) 4,704 ÷ 14 = 336

57) 1,820 ÷ 13 = 140

58) 7,062 ÷ 33 = 214

59) 8,671 ÷ 29 = 299

Long Division: Practice (continued)

Part 6 More dividing (continued)

60) 16,779 ÷ 51 = 329

61) 5,720 ÷ 40 = 143

62) 10,972 ÷ 52 = 211

63) 23,247 ÷ 63 = 369

64) 12,825 ÷ 45 = 285

65) 21,560 ÷ 55 = 392

Part 7 Remember? Problems from earlier packets.

66) $\frac{3}{4} \div \frac{1}{8} = 6$

67) 35% of 60 = 21

68) 72 pages in 4 days = 18 pages per day

69) 13 : 6 = 39 : 18

70) 3.2 × 5 = 16

71) $\frac{7}{10} - \frac{1}{4} = \frac{9}{20}$

Long Division: Practice (continued)

Part 8 Word problems Divide, multiply, take away, bring down.

72) A school has 1,248 books on 16 shelves. Each shelf holds the same. How many books on a shelf?

Number sentence 1,248 ÷ 16 = 78

Answer 78 books

Draw or show your work

73) A farm packs 3,456 eggs in cartons of 12. How many cartons?

Number sentence 3,456 ÷ 12 = 288

Answer 288 cartons

Draw or show your work

74) A trip of 2,310 miles takes 42 hours. How many miles an hour is that?

Number sentence 2,310 ÷ 42 = 55

Answer 55 miles

Draw or show your work

75) 1,000 stickers are shared by 24 kids. How many does each kid get?

Number sentence 1,000 ÷ 24 = 41 R 16

Answer 41 stickers

Draw or show your work

76) Challenge: 1,000 people go to a game. A bus holds 45. How many buses are needed?

Number sentence 1,000 ÷ 45 = 22 R 10, so one more

Answer 23 buses

Draw or show your work

Check-In: Long Division

Show what you know. No help this time.

Mastery: 7 of 8 (80%)

Part 1 Long division

1) $26 \overline{)6{,}552}$ = 252

2) $8 \overline{)2{,}592}$ = 324

Part 2 Divide

3) 5,542 ÷ 17 = 326

4) 2,982 ÷ 14 = 213

5) 4,469 ÷ 41 = 109

6) 6,020 ÷ 35 = 172

Part 3 What is left?

7) 7,250 ÷ 34 = 213 R 8

8) 3,000 ÷ 29 = 103 R 13